

# Classroom Supply Lists

final proj. o ironhack DSML course June'26

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2026/08/14

# Problem

unstructured documents from school demand specific items of classroom supply:

notebooks, binders, pens etc.

diverse implementation of specific needs built by personal taste of teachers

unstructured documents from school demand specific items of classroom supply:  
notebooks, binders, pens etc.  
diverse implementation of specific needs built by personal taste of teachers

[illegible]

# Problem

Parent's job has worsened.

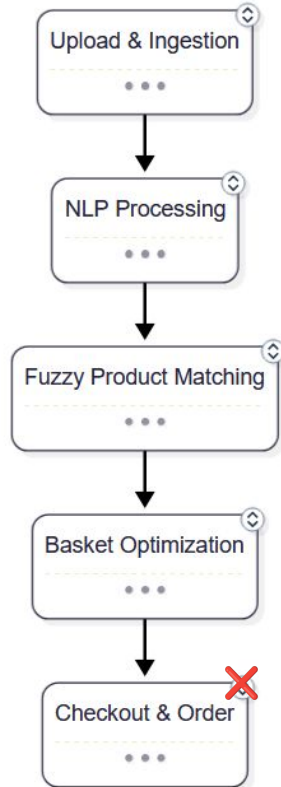
A local supplier is profiting from unnecessary expenses related to the delivery of classroom supplies.

But there is a potential of minimizing workload:

Digitalization through data science and machine learning ...

# Solution: OCR School Supply Scanner & Basket Optimization

workflow:



<https://mgrillo.de> ocr

preprocessing  
(highlighting)

similarity score

pretrained model for  
further optimization of  
basket (grade level or  
teaching subject)

## Upload & Ingestion

<https://mgrillo.de> ocr

preprocessing (highlighting)

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pretrained model for further  
optimization of basket (grade level  
or teaching subject)

- ibm-granite-d-258M (huggingface)
  - 300M params. I needed 800 tokens per page. 30 min on T1.
  - + result: “6 Hefte” (notebook), “DIN A+”
- ←—————→
- tesseract ocr
  - open source software, way faster, highly configurable
  - + result: “6 Hefte” (notebook), “DIN At”

6 Hefte, DIN A4, ohne Rand, Lineatur 3

# NLP Processing

<https://mgrillo.de> ocr

preprocessing  
(highlighting)

similarity score

pretrained model for further  
optimization of basket (grade level  
or teaching subject)

- tesseract.config/\_words.txt
- RegEx (unreadable char.)
- 'Envelope, binder, folder, notebook'-logic hard coded in text\_optimizer class
- quantities vs. product spec numbers

6 Hefte, DIN A4, ohne Rand, Lineatur 3

# NLP similarity score & fuzzy product matching

<https://mgrillo.de> ocr

preprocessing (highlighting)

similarity score

pretrained model for further  
optimization of basket (grade level  
or teaching subject)

- embeddings of all text
- matching to **products.csv**  
(description, label, class)
- difflib.SeqMatcher / RapidFuzz (no semantic dist!)
- Resources for sentence-transformers  
(500MB RAM <https://mgrillo.de> No space left on device)

6 Hefte, DIN A4, ohne Rand, Lineatur 3





# Erkannte Produkte verifizieren

## Erkannter Text

Materialliste Klasse 7

Fach Material

Deutsch 2x Heft DIN A4 liniert mit Rand



roter DIN A4 Einband



roter Schnellhefter



Englisch 1x Heft DIN A4 liniert mit Rand



oranger DIN A4 Einband



oranger Schnellhefter



Vokabelheft



oranger Einband für das Vokabelheft



Mathe 2x Heft DIN A4 kariert mit Rand



## Katalog-Produkt

## Menge



Heft DIN A5 kariert, Lineatur 7 (16 Blatt) (Brunnen) - 0.99 €



1



Schnellhefter A4, Kunststoff schwarz (Durable) - 0.50 €



1

Heft DIN A4 liniert, Lineatur 25 mit Rand (16 Blatt) (Brunnen) - 0.99 €



1



Umschlag DIN A4, rot (Brunnen) - 0.45 €



2



Schnellhefter A4, Kunststoff blau (Durable) - 0.50 €



1

Heft DIN A4 liniert, Lineatur 25 mit Rand (16 Blatt) (Brunnen) - 0.99 €



1

Umschlag DIN A4, rot (Brunnen) - 0.45 €



1

Schnellhefter A4, Kunststoff blau (Durable) - 0.50 €



1

Vokabelheft DIN A5, Lineatur 53 (32 Blatt, zweispaltig/dreispaltig) (Oxford) - 1.29 €



1

Heft DIN A5 blanko, Lineatur 6 (16 Blatt) (Brunnen) - 0.99 €



1

Heft DIN A4 kariert, Lineatur 26 mit Rand (16 Blatt) (Brunnen) - 0.99 €



1



## basket optimization

<https://mgrillo.de> ocr

preprocessing (highlighting)

similarity score

pretrained model for  
further optimization of  
basket (grade level or  
teaching subject)

- *grade level* made no sense  
due to  
preceding poor decisions and results  
+ lack of proper training data
- *teaching subject* could aggregate on similarity  
score (open todo)

NB: I struggled significantly with integrating the  
cluster-based knowledge into the workflow.

6 Hefte, DIN A4, ohne Rand, Lineatur 3

# Thank you very much!



mgrillo.de



Photo

PDF

Cart

Pull

## Analyze PDF Order List

Upload the school's digital PDF supply list to extract items automatically.

Upload PDF file...



200MB per file • PDF



## Your Current Cart

The cart is empty.